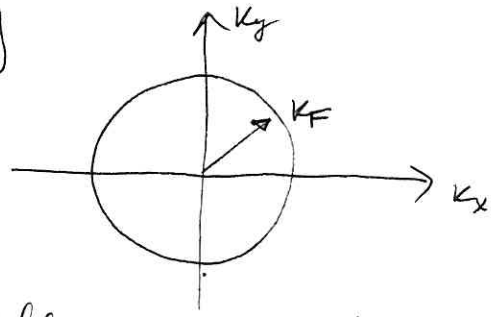
P3



$\underline{\underline{T=0}}$

(a) All states with $k < k_F$ are occupied
states with $k > k_F$ are empty

$\underline{d=2}$

$\Rightarrow$ for N electrons : $N = 2 \sum_k \langle n_k \rangle$

$d=2 \Rightarrow \vec{k} = (k_x, k_y)$ spin ↑ or ↓ degeneracy.

$$N = 2 \sum_k \langle n_k \rangle = 2 \underbrace{\left(\frac{L}{2\pi}\right)^2}_{\text{density of states}} \int d^2\vec{k}\,(1) =$$

$$= \frac{2L^2}{(2\pi)^2} \int_0^{k_F} dk \cdot k \cdot 2\pi = \frac{L^2}{\pi} \frac{k_F^2}{2} = \frac{L^2}{2\pi} k_F^2.$$

$$\Rightarrow \quad k_F = \left(2\pi \frac{N}{L^2}\right)^{1/2} \qquad \frac{N}{L^2} = \frac{N}{\text{area}} = n = \text{concentration.}$$

$$\varepsilon_F = \frac{\hbar^2}{2m} k_F^2 = \frac{\hbar^2}{2m} 2\pi \frac{N}{L^2} = \frac{\pi \hbar^2 N}{m L^2}$$

(b) $\underline{d=1}$ $\quad N = 2 \sum_k \langle n_k \rangle = 2 \frac{L}{2\pi} \int_{-k_F}^{k_F} dk = \frac{L}{\pi} 2 k_F$

↓
k has just one component!

$$k_F = \frac{\pi N}{2L} = \frac{\pi}{2} n \quad \left(n = \frac{N}{L} = \text{one-dimensional density}\right)$$

$$\varepsilon_F = \frac{\hbar^2}{2m} k_F^2 = \frac{\hbar^2}{2m} \left(\frac{\pi n}{2}\right)^2$$

Problem 5.35 = P4

(a)

$$V = \frac{4}{3}\pi R^3, \quad \text{so } E = \frac{\hbar^2(3\pi^2 Nq)^{5/3}}{10\pi^2 m}\left(\frac{4}{3}\pi R^3\right)^{-2/3} = \boxed{\frac{2\hbar^2}{15\pi m R^2}\left(\frac{9}{4}\pi N q\right)^{5/3}}.$$

(b) Imagine building up a sphere by layers. When it has reached mass m, and radius r, the work necessary to bring in the next increment dm is: $dW = -(Gm/r)\,dm$. In terms of the mass density ρ, $m = \frac{4}{3}\pi r^3\rho$, and $dm = 4\pi r^2 dr\rho$, where dr is the resulting increase in radius. Thus:

$$dW = -G\frac{4}{3}\pi r^3\rho\, 4\pi r^2\rho\frac{dr}{r} = -\frac{16\pi^2}{3}\rho^2 G r^4 dr,$$

and the *total* energy of a sphere of radius R is therefore

$$E_{\text{grav}} = -\frac{16\pi^2}{3}\rho^2 G\int_0^R r^4\,dr = -\frac{16\pi^2\rho^2 R^5}{15}G. \quad \text{But } \rho = \frac{NM}{4/3\pi R^3}, \text{ so}$$

$$E_{\text{grav}} = -\frac{16\pi^2 R^5}{15}G\frac{9N^2M^2}{16\pi^2 R^6} = \boxed{-\frac{3}{5}G\frac{N^2M^2}{R}}.$$

(c)

$$E_{\text{tot}} = \frac{A}{R^2} - \frac{B}{R}, \quad \text{where } A \equiv \frac{2\hbar^2}{15\pi m}\left(\frac{9}{4}\pi N q\right)^{5/3} \text{ and } \quad B \equiv \frac{3}{5}GN^2M^2.$$

$$\frac{dE_{\text{tot}}}{dR} = -\frac{2A}{R^3} + \frac{B}{R^2} = 0 \Rightarrow 2A = BR, \quad \text{so } R = \frac{2A}{B} = \frac{4\hbar}{15\pi m}\left(\frac{9}{4}\pi N q\right)^{5/3}\frac{5}{3GN^2M^2}.$$

$$R = \left[\left(\frac{4}{9\pi}\right)\left(\frac{9\pi}{4}\right)^{5/3}\right]\left(\frac{N^{5/3}}{N^2}\right)\frac{\hbar^2}{GmM^2}q^{5/3} = \boxed{\left(\frac{9\pi}{4}\right)^{2/3}\frac{\hbar^2}{GmM^2}\frac{q^{5/3}}{N^{1/3}}}.$$

$$R = \left(\frac{9\pi}{4}\right)^{2/3}\frac{(1.055\times10^{-34}\,\text{J}\cdot\text{s})^2(1/2)^{5/3}}{(6.673\times10^{-11}\,\text{Nm}^2/\text{kg}^2)(9.109\times10^{-31}\,\text{kg})(1.674\times10^{-27}\,\text{kg})^2}N^{-1/3}$$
$$= \boxed{(7.58\times10^{25}\,\text{m})N^{-1/3}.}$$

(d) Mass of sun: 1.989×10^{30} kg, so $N = \dfrac{1.989\times10^{30}}{1.674\times10^{-27}} = 1.188\times10^{57}$; $N^{-1/3} = 9.44\times10^{-20}$.

$R = (7.58\times10^{25})(9.44\times10^{-20})\,\text{m} = \boxed{7.16\times10^6\,\text{m}}$ (slightly larger than the earth).

(e)

From Eq. 5.43: $E_F = \dfrac{\hbar^2}{2m}\left(3\pi^2\dfrac{Nq}{4/3\pi R^3}\right)^{2/3} = \dfrac{\hbar^2}{2mR^2}\left(\dfrac{9\pi}{4}Nq\right)^{2/3}$. Numerically:

$$E_F = \frac{(1.055\times10^{-34}\,\text{J}\cdot\text{s})^2}{2(9.109\times10^{-31}\,\text{kg})(7.16\times10^6\,\text{m})^2}\left[\frac{9\pi}{4}(1.188\times10^{57})\frac{1}{2}\right]^{2/3} = 3.102\times10^{-14}\,\text{J},$$

or, in electron volts: $E_F = \dfrac{3.102\times10^{-14}}{1.602\times10^{-19}}\,\text{eV} = \boxed{1.94\times10^5\,\text{eV}.}$

$E_{\text{rest}} = mc^2 = 5.11\times10^5$ eV, so the Fermi energy (which is the energy of the most energetic electrons) is comparable to the rest energy, so they are getting fairly relativistic.